

C++

PENG

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1 Variables and Basic Types

1.1 Primitive Built-in Types

1.1.1 Arithmetic Types

Definition 1.1 (Primitive builtin types).

- *arithmetic types*:
 - *integral types*
 - * *characters*: `char`, `std::string`
 - * *integers*: `int`: signed or unsigned (≥ 0)
 - * *boolean values*: `bool`
 - *floating-point types*: `double`
- *void*

Remark 1.2. Use `double` for *floating-point computations*.

1.1.2 Type Conversions

Example 1.3 (Type conversions).

```
bool b = 20; // 0 for false and nonzero values for true
int i = b;   // true for 1 and false for 0
```

Example 1.4 (Signed and unsigned).

```
unsigned u = -1; // if 32-bit ints, 4294967295
int a = -1;
unsigned b = 1;
a * b // int value is converted to unsigned before addition
```

1.1.3 Literals

Definition 1.5 (Literals).

Integer literals:

- *decimal*
- *octal*: 0-
- *hexadecimal*: 0x-
- *suffix*:
 - *u*: *unsigned*
 - *L*: *long*

Floating-point literals (double by default):

- *decimal point*
- *E*: 1E-2
- *suffix*: f/F: *float*

character and string literals: "Hello" = "Hello\0"

Definition 1.6 (Escape sequences).

- *newline*: \n
- *horizontal tab*: \t
- *null*: \0

1.2 Variables

1.2.1 Variable Definitions

Remark 1.7.

- *Variables defined outside any function body are initialized to 0.*
- *Initialize every object of built-in type.*
- `std::cin >> int i;` *error, should be* `int i; std::cin >> i;`.

1.2.2 Variable Declarations and Definitions

File A define the variable `int ix;`. Other files: `extern int ix;`.

All variables must be declared before being used.